

MODULE 01 · DOMAIN 1 - CONTEXT AND RECONCILIATION

The Context Package

Why the demo worked on their database and fails on yours.

DOMAIN 1 of 5 MODULE 1 of 11 PROMPTS 2

What this course is, and is not

- ◆ **It is about judgement.** Thirty two prompts, and why each is shaped the way it is.
- ◆ **It is warehouse work.** Mapping documents, stored procedures, grain, joins, drift.
- ◆ **It is not a tool tutorial.** Nothing here depends on which model you use.
- ◆ **Everything is copy and paste.** Every prompt appears in full, on screen.

WHY IT IS BUILT THIS WAY

The prompts will age. The reasons they are shaped this way will not.

Accuracy does not decline as databases get realistic. It **collapses**.

The same task, four benchmarks

BENCHMARK	ACCURACY	WHAT CHANGED
Spider	around 91%	Small, clean, tidy schemas
BIRD	around 77%	Larger schemas, messier values
Spider 2.0	roughly 17 to 21%	Real enterprise schemas
BEAVER	close to zero	Real warehouse, real queries

THE POINT

Same task, same models. Only the databases changed.

What actually drives that

- ◆ **Column counts.** Benchmark schemas average tens of columns. Yours run to hundreds.
- ◆ **Join depth.** Benchmark queries need about one join. Under one percent need four.
- ◆ **Naming.** Benchmark columns are called what they are. Yours are called MTART.
- ◆ **None of that is a model problem.** It is a context problem, and you control it.

THE POINT

The gap between the demo and your job is almost entirely the schema you did not hand over.

Two databases

THE DEMO

Tidy

- Around 28 columns
- Column names describe the data
- One join, usually zero
- One meaning per column

YOUR WAREHOUSE

Real

- Hundreds of columns
- MATNR, WERKS, LGORT, BWART
- Six joins across four schemas
- One column meaning three things

THE TIEBREAKER

You cannot make your warehouse tidy. You can make the part you hand over legible.

Assemble the context. Do not describe it.

Three things, in this order

- ◆ **The DDL, verbatim.** Create table statements, pasted. Not a summary.
- ◆ **Five sample rows per table.** Real values, or realistic ones.
- ◆ **The lookup contents.** If a column holds codes, paste the code table.
- ◆ **Then the question.** Last, not first.

WHY DDL AND NOT PROSE

The research calls it schema linking. The model has to connect your words to exact column names.

Two ways to describe the same table

Both are accurate. Only one is usable.

```
01 -- What people paste:
02 "The parts table has a part number, a site, stock levels,
03  and a planning type field."
04
05 -- What actually works:
06 CREATE TABLE pmats (
07     partno  VARCHAR(40), site   VARCHAR(10),
08     stloc   VARCHAR(10), plntyp VARCHAR(6),
09     minqty  NUMERIC(18,6), maxqty NUMERIC(18,6) );
10
11 -- The prose never mentions stloc. It is in every filter.
```

WHAT IT COST

The summary dropped the stock location column. Every subsequent query would ignore it.

Why sample rows matter more than they look

- ◆ **They show what a code looks like.** Is the status 'A' or 'ACTIVE' or 1?
- ◆ **They show the real types.** A date column that is actually a string.
- ◆ **They show null behaviour.** Real nulls, or empty strings pretending to be nulls.
- ◆ **Half of all wrong output** traces to an assumption about values, not structure.

THE POINT

Five rows per table. It takes two minutes and it removes a whole category of error.

What you actually type

The framing above the paste is short. The paste is the work.

Below is the schema I am working with. Use only these tables and columns. Do not assume any column that is not listed.

1. CREATE TABLE statements, verbatim
2. Five sample rows per table
3. Contents of every lookup table referenced

If you need a column that is not here, stop and ask. Do not substitute a column that looks similar.

I will give you the actual task in my next message. For now, tell me what you understand this schema to represent.

THE LINE THAT CATCHES PROBLEMS EARLY

A cheap test. A wrong summary means everything after it would be wrong.

Before you ask for anything, ask what is
missing.

What the gap audit finds

- ◆ **Lookup tables you forgot.** A code column with no code table pasted.
- ◆ **Relationships you assumed.** Two tables that must join, with no obvious key.
- ◆ **Undefined values.** A status column whose codes appear nowhere.
- ◆ **Cheaper now than later.** Two minutes here, or an afternoon of wrong output.

THE POINT

It is asking what it would need to guess at. Guesses are what you are trying to prevent.

What you actually type

Run it after the schema, before the task.

```
Here is the task I am about to give you: <one line summary>
```

```
Do not attempt it. Instead, tell me what is missing from the  
schema I pasted that you would need in order to do it.
```

```
Specifically:
```

- Any column referenced by the task but absent
- Any code or status column with no lookup provided
- Any join you would need where the key is not obvious
- Anything you would have to assume about the data

```
If nothing is missing, say so. Do not invent gaps.
```

THE SHAPE

Give it the task in one line so it knows what to check for, then forbid it from starting.

What it returned on a real schema

Nine tables pasted, one task described

- ◆ **Two missing lookups.** Movement type and part type codes, both filtered on, neither defined.
- ◆ **One ambiguous join.** Two tables sharing three columns. Which are the key?
- ◆ **Four assumptions** it would have made silently, now listed.
- ◆ **Total time to fix.** About six minutes of pasting.

THE RESULT

Every one of those would have become a confident, invisible decision.

Your fact table has 140 columns. The task involves eight of them. You are about to paste the schema.

Pasting all 140 uses a lot of the context window and includes a great deal that is irrelevant.

What do you paste?

- ☐ A All 140. More context is safer
- ☐ B The 8 you need. Everything else is noise
- ☐ C The 8, plus keys, plus anything named similarly to the 8
- ☐ D A summary of all 140, plus full DDL for the 8

◆ PAUSE AND DECIDE

What do you paste?

- A

 All 140
More context is not reliably better. Quality degrades as irrelevant material accumulates.
- B

 Just the 8
Too tight. It cannot join without keys, and it cannot warn you about a column you overlooked.
- C

 The 8, keys, and near-names
Enough to work with, and near-names let it flag a column you may have picked wrongly.
- D

 Summary plus DDL
The summary of 132 columns adds length without precision. That is the worst of both.

THE SIGNAL

Similar names matter. If there is a `total_qty` and a `total_qty_net`, you want to be told.

You paste the schema and ask what it represents. The summary comes back and describes your usage fact table as a snapshot of current stock levels.

It is not. It is a rolling twelve month consumption total.

What does this tell you?

- ☐ A The model is unreliable. Proceed with caution
- ☐ B The table name and columns genuinely suggest a snapshot
- ☐ C The sample rows were not representative
- ☐ D Correct it and move on. It is only a summary

◆ PAUSE AND DECIDE

What does this tell you?

- A

 Model unreliable
It read your schema and reached a reasonable conclusion. That is information about the schema.
- B

 The schema suggests it
The naming genuinely misleads. A new colleague would make the same mistake next month.
- C

 Bad sample rows
Worth checking, and rolling totals and snapshots look identical in five rows.
- D

 Correct it, move on
Fixes this session and leaves the trap in place for everyone else.

THE SIGNAL

The check just told you something true about your warehouse, not about the model.

The gap audit says it needs the lookup table for a status column in order to write the filter correctly.

You know the two codes that matter. Typing them takes ten seconds. Exporting the lookup table takes twenty minutes.

What do you do?

- ☐ A Type the two codes you know. It is what the filter needs
- ☐ B Export the full lookup table
- ☐ C Type the two codes, and note that the list is unverified
- ☐ D Ask the model to infer the codes from the sample rows

◆ PAUSE AND DECIDE

What do you do?

- A

 Type the two codes
Fine for the query and it launders your memory into something that reads as verified.
- B

 Export the full table
Thorough, and twenty minutes is a real cost for a filter you already understand.
- C

 Type them, mark unverified
Keeps the speed and keeps the uncertainty attached to the output.
- D

 Infer from samples
Five rows cannot tell you which codes exist, only which happened to appear.

THE SIGNAL

The pattern recurs all course. Use what you know, and record that you were the source.

The gap audit comes back saying nothing is missing. Your schema paste covers everything needed.

This is the first time you have run it and the task is a six join aggregation across four schemas.

How do you read that?

- ☐ A Good. The context package was complete
- ☐ B Suspicious. Push on the joins specifically
- ☐ C Meaningless. It always says that
- ☐ D Re-run it with the task described in more detail

◆ PAUSE AND DECIDE

How do you read that?

- A

 Package was complete
Possible on a simple task. On six joins across four schemas it is worth thirty seconds of doubt.
- B

 Push on the joins
Ask it to state each join key explicitly. Ambiguity surfaces there before anywhere else.
- C

 It always says that
It does not. The prompt forbids inventing gaps, and it reports them when they exist.
- D

 More task detail
Reasonable and vague. B tests the specific thing most likely to be wrong.

THE SIGNAL

Same shape as a grain declaration with no fan-out risk. A clean answer on a complex task earns a push.

Take this to work tomorrow

- ◆ Paste DDL, never a description. Five sample rows per table.
- ◆ Include lookup contents for every code column you filter on.
- ◆ Ask what the schema represents before you ask for anything else.
- ◆ Run the gap audit. Six minutes of pasting beats an afternoon of plausible output.

THE PAYOFF

None of this is prompting. It is preparation, and it decides everything downstream.

You have the schema. Next: the documents that describe it, and why they are **wrong**.